

patients showed toxicities $\geq$ G3. Median FU time 10 months (range 1-45 months). We observed Complete/Partial response (CR and PR, respectively) in 82.4% of the cases, with 30/51 CR (58.9%) and 12/51 (23.5%) PR. Progressive disease was reported in 13 patients (7 local progression, 1 local and systemic, 5 systemic).

Conclusion: These preliminary results showed that SBRT boost was feasible and well tolerated with encouraging results in terms of local control. Our data seem to indicate that this kind of therapy is a valid therapeutic option in LACC patients who cannot be submitted to endocavitary brachytherapy. At the best of our knowledge this is the largest study reported in English literature using a Robotic Arm LINAC.

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2820

Optic Nerve Dose and Long-Term Visual Outcomes for Uveal Melanoma Patients Treated with Eye Plaque Brachytherapy

J. Pasetsky,¹ L. Yeager,² E. Tsamis,² Y. Wang,² B.P. Marr,² and I. Deutsch¹; ¹Department of Radiation Oncology, Columbia University Irving Medical Center, New York, NY; ²Department of Ophthalmology, Columbia University Irving Medical Center, New York, NY

Purpose/Objective(s): Eye plaque brachytherapy is the most common and effective treatment for Uveal Melanoma, allowing patients to avoid enucleation while providing excellent long term oncologic outcomes. Unfortunately, many patients will have decreased visual acuity from treatment. Our study aimed to determine if a relationship exists between long term visual acuity and dose along the optic nerve (ON), with the goal of improving treatment planning and optimizing long term visual outcomes in patients treated with eye plaque brachytherapy.

Materials/Methods: All patients treated at our institution for Uveal Melanoma with I-125 eye plaque brachytherapy to 85 Gy were eligible for our study. ON dosimetry was available starting in 2018 and dose to the ON was calculated at the optic disc and up to 3mm distally towards the brain (1 mm increments). Patients with this data were eligible if they had at least 12 months of follow up after treatment to allow maturation of possible visual deficits. 20/200 vision or worse is considered legal blindness. Univariate Cox proportional hazard models were used to evaluate the effect of radiation dose up to 3 mm along the ON on the risk of blindness.

Results: 118 patients, with 119 total eyes treated, had data available for visual acuity and dose to the optic nerve, of which 106 eyes at least 1 year of follow up for visual acuity. Median follow up was 2.55 years. 43 total eyes were permanently blinded, 8 of which were blind pre-treatment. Of the 35 patients blinded by treatment (36%), median time to blindness was 1.3 years. Mean dose at the optic disc for the 35 blinded eyes and the 63 un-blinded eyes was 57.35 Gy and 39.02 Gy respectively. Mean dose at 1mm, 2 mm and 3 mm along the optic nerve was 25.29 Gy, 14.87 Gy and 10.14 Gy in the blinded eyes, respectively, while only 18.60 Gy, 11.22 Gy and 9.47 Gy in the unblinded eyes, respectively. Univariate analysis of dose at 0 mm, 1 mm, 2 mm and 3 mm leading to blindness revealed Hazard Ratios of 1.01, 1.03, 1.07 and 1.0 with p-values <0.001 for each, other than at 3 mm which had a p-value of .621.

Conclusion: Dose at the optic disc and along the ON may be correlated to long term blindness in patients with Uveal Melanoma treated with eye plaque brachytherapy. Additional dosimetric analysis can shed light on optimizing dose constraints to maintain long term visual acuity and help counsel patients on risks of legal blindness.

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2821

Outcomes in Patients with Benign Tumors of the Head and Neck Treated with LINAC-Based Stereotactic Body Radiation Therapy

A. Patel,¹ A. Lee,² H. Wang,³ X.A. Wang,⁴ A.C. Moreno,² J. Reddy,² M.T. Spiotto,² D.I. Rosenthal,² S.M. Raza,⁵ F. DeMonte,⁵ P.W. Gidley,⁶ M.E. Nader,⁶ S.Y. Su,⁶ E.Y. Hanna,⁶ A.S. Garden,² and J. Phan²; ¹Paul L. Foster School of Medicine, Texas Tech Health Sciences Center El Paso, El Paso, TX, ²Department of Radiation Oncology, University of Texas MD Anderson Cancer Center, Houston, TX, ³Department of Radiation Physics, The University of Texas MD Anderson Cancer Center, Houston, TX, ⁴Department of Radiation Physics, University of Texas MD Anderson Cancer Center, Houston, TX, ⁵Department of Neurosurgery, University of Texas MD Anderson Cancer Center, Houston, TX, ⁶Department of Head and Neck Surgery, University of Texas MD Anderson Cancer Center, Houston, TX

Purpose/Objective(s): Benign tumors of the head and neck involving cranial nerves (CN) are rare entities that are difficult to treat given their proximity to critical structures. While slow growing, surgical resection can be associated with high morbidity rates. We present our institutional experience in treating benign head and neck tumors with LINAC-based stereotactic body radiation therapy (SBRT).

Materials/Methods: Patients enrolled in our prospective SBRT registry between 2017-2023 were included in this analysis. All patients were evaluated for surgery, with radiation therapy considered primarily due to the risk of potential CN injury. Tumors that were ineligible for single-session non-invasive stereotactic radiosurgery due to factors such as large volume or location below C1, but measured <60 cc, were considered for SBRT. Outcomes of interest included local control (LC), overall survival (OS), toxicity and symptomatology. Local control was defined as radiographic evidence of lack of progression.

Results: The cohort consisted of 30 patients with a median age of 51 years. Among them, 23 cases were paragangliomas (PGL), distributed as follows: jugulotympanic PGL (n=7), jugular PGL (n=13), and vagal PGL (n=3). Additionally, 7 cases were non-vestibular schwannomas, involving the hypoglossal nerve (CN XII, n=4) and vagus nerve (CN X, n=3). The primary indication for treatment was tumor growth or symptomatic progression. At the time of SBRT, 29 patients had gross disease and 1 received adjuvant treatment after surgical resection. Among those with gross disease, 5 patients had prior treatment before SBRT: 4 underwent surgery, and one received prior fractionated radiation. The median target volume was 13.5 cm³ (range: 1.32–48.5 cm³), and the median prescribed dose was 27 Gy in 3 fractions (range: 21–36 Gy in 3–5 fractions), with a median prescription coverage of 96% (range: 86–100%). With a median follow-up of 36 months, all 30 patients treated for gross disease had radiographically stable tumors, while the patient treated adjuvantly remained disease-free. Hearing loss, tinnitus, and dizziness were the most commonly reported pre-treatment symptoms in patients with paragangliomas, while dysphagia and hoarseness were the most frequent symptoms among those with schwannomas. Seventeen patients (55%) reported an improvement in symptoms and 14 (45%) reported no worsening symptoms. Following treatment, there was 1 grade 1 tinnitus and 1 grade 2 nausea, but no grade 3 or higher toxicities were observed.

Conclusion: In the largest study to date, LINAC-based SBRT for benign head and neck tumors with cranial nerve involvement demonstrated excellent tolerability with minimal toxicity. With a median follow up of 36 months, most patients experienced stable or improved pre-treatment symptoms, with optimal local control. These findings support LINAC-based SBRT as an effective treatment option for benign head and neck tumors arising from jugular foramen or involving cranial nerves. Longer follow up is needed to validate these outcomes.

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